

Wednesday Warm Up: Unit 7: Forces V

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1 Memory Bank

- $\hat{i} \cdot \hat{i} = 1, \hat{i} \cdot \hat{j} = 0 \dots$ The dot product between unit vectors.
- $\vec{u} \cdot \vec{v} = uv \cos \theta \dots$ The dot product of two vectors is the product of the magnitudes times the cosine of the angle between them.
- $\hat{i} \times \hat{j} = \hat{k}, \hat{k} \times \hat{i} = \hat{j}, \hat{j} \times \hat{k} = \hat{i} \dots$ The *cross-product* between unit vectors.
- $\hat{j} \times \hat{i} = -\hat{k}, \hat{i} \times \hat{k} = -\hat{j}, \hat{k} \times \hat{j} = -\hat{i} \dots$ When the unit vectors are not in alphabetical order, there is a minus sign.
- $\tau = \vec{r} \times \vec{F} \dots$ Definition of torque.
- $|\tau| = rF \sin \theta \dots$ The magnitude of the torque is the product of r and F , times the sine of the angle between them. The distance r is between the force and the pivot point.
- Torque is the angular version of force. It is the result of a force F , separated from a pivot by a distance r , that rotates the system about the pivot. The angle between the force F and the distance r is θ .
- **For systems in static equilibrium, (1) the net force is zero, and (2) the net torque is zero.**

2 Dot Product, and Cross Product

1. Calculate the dot product:

- $\vec{a} = 2\hat{i} - 2\hat{j}, \vec{b} = -4\hat{i} + 4\hat{j}.$
- $\vec{a} = 2\hat{i}, \vec{b} = -4\hat{k}.$
- $\vec{a} = 2\hat{i} + 2\hat{j}, \vec{b} = -2\hat{i} - 2\hat{j}.$

2. Calculate the cross product:

- $\vec{a} = 2\hat{i} - 2\hat{j}, \vec{b} = -4\hat{i} + 4\hat{j}.$
- $\vec{a} = 2\hat{i}, \vec{b} = -4\hat{k}.$
- $\vec{a} = 2\hat{i} + 2\hat{j}, \vec{b} = -2\hat{i} - 2\hat{j}.$

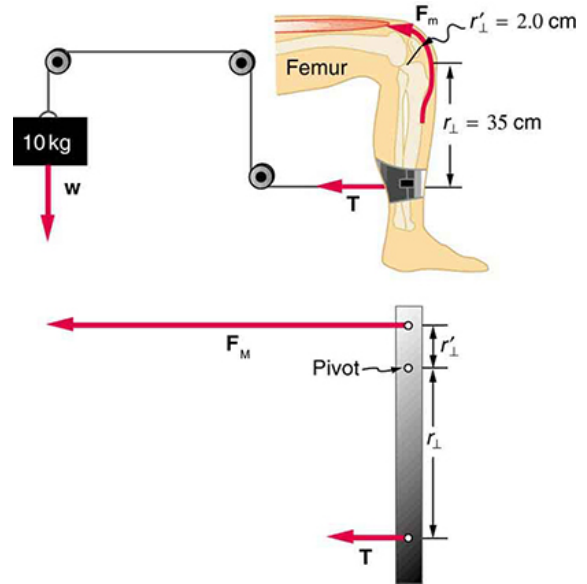


Figure 1: An exercise machine for the quadriceps requires torque.

3 Torque

1. A device for exercising the upper leg muscle is shown in Fig. 1, together with a schematic representation of an equivalent lever system. Calculate the magnitude of the force exerted by the upper leg muscle to lift the mass at a constant speed.